

Product Reference Manual (V1.0)

Description

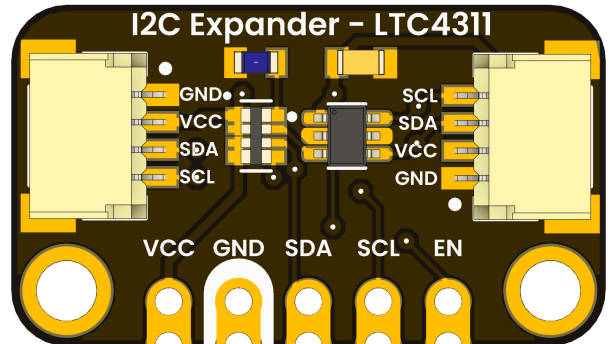
The I2C LTC4311 Bus Extender Module is designed to enhance the I2C communication range and reliability in your embedded systems. It utilizes the LTC4311 bus extender IC to boost I2C signals, allowing for longer cable lengths and improved noise immunity. This module is ideal for applications requiring robust I2C communication over extended distances.

The device monitors the SDA and SCL lines and automatically injects current during rising transitions, resulting in cleaner signal edges and improved communication reliability. This allows the I²C bus to operate at standard and fast-mode speeds without requiring changes to pull-up resistors, firmware, or system configuration.

The module enables stable operation in applications where conventional I²C communication would be limited by signal degradation.

DevLab format compatibility.

Simplicity and compatibility are the core principles of the **DevLab form factor**. This standard defines a **compact and communication-optimized board layout**, ensuring straightforward connection and interoperability among DevLab module



Key Features

- **I²C Interface** — Simplified communication with most microcontrollers and SBCs.
- Extended I²C communication over long cable length
- Active rise-time acceleration on SDA and SCL lines
- Automatic operation without configuration or firmware changes
- Wide operating voltage range from 1.6 V to 5.5 V
- Support for I²C communication speeds up to 400 kHz
- Improved noise margin and signal stability
- Low-power standby mode
- High ESD robustness

Hardware Features

- LTC4311 active I²C terminator integrated circuit
- Dual I²C connectors for inline bus integration
- Standard DevLab pin header (3V3, GND, SDA, SCL, EN)
- Enable (EN) control pin for device activation
- Compact PCB optimized for signal integrity
- Compatibility with STEMMA QT / Qwiic ecosystems
- Passive insertion into existing I²C bus without modification

Applications

- Long-distance I²C communication systems
- Sensor networks distributed over extended wiring
- Industrial and embedded monitoring systems
- Prototyping platforms with multiple I²C devices
- Robotics and automation systems
- Data acquisition systems with remote sensors
- Educational and development environments requiring robust I²C connectivity

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1 The Board

1.1 Accessories

2 Ratings

2.1 Recommended Operating Conditions

Symbol	Description	Min	Typ	Max	Unit
VCC	Supply Voltage Range	1.6	—	5.5	V
fSCL	I ² C Clock Frequency	—	—	400	kHz
ISHDN	Shutdown Supply Current	—	—	5	μA
ESD	ESD Protection (HBM)	—	—	±8	kV
trise	Rise Time Improvement	—	Enhanced	—	—
NM_L	Low State Noise Margin	—	Improved	—	—
IDRIVE	Pull-up Assist Current	—	Active	—	—
ILOAD	Bus Load (Disabled)	—	0	—	—
MODE	Low Power Mode	Auto	—	—	—
MULTI	Multi-device Support	Yes	—	—	—

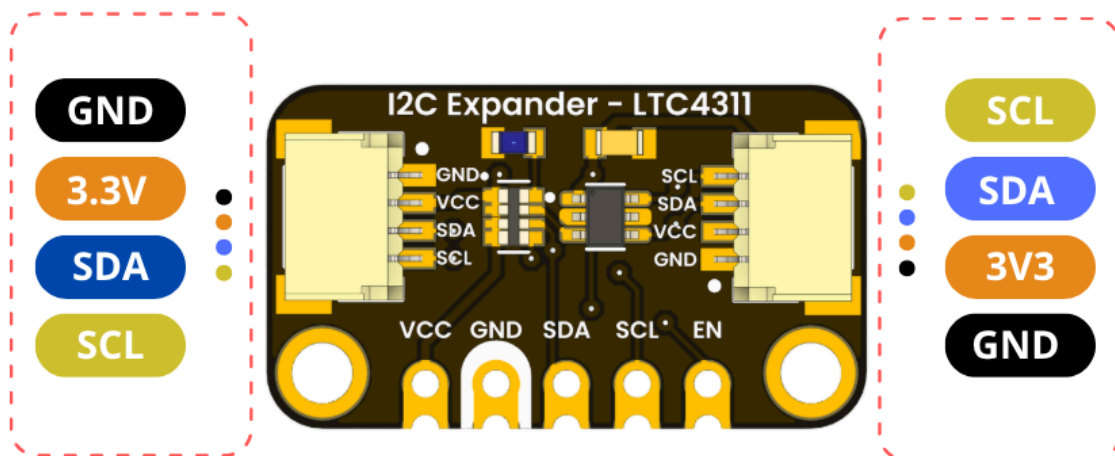
Notes

- *trise / NM_L / IDRIVE* → qualitative (no fixed numeric value in datasheet summary)
- *ILOAD = 0* → meaning does not load the bus when disabled
- *MODE = Auto* → automatic standby detection
- *MULTI = Yes* → supports multiple I²C devices reliably

3 Functional Overview

Signal Conditioning and Rise Time Acceleration

In conventional I²C systems, signal rise time is determined by the interaction between pull-up resistors and total bus capacitance. As cable length increases or additional devices are connected, the effective capacitance rises, resulting in slower signal transitions. This degradation can lead to distorted waveforms, reduced noise margins, and unreliable communication.



The LTC4311 improves this behavior by actively monitoring the SDA and SCL lines and accelerating rising edges through controlled current injection. This mechanism enhances signal transitions without requiring modification of external pull-up resistors. As a result, slow, rounded waveforms are reshaped into sharper transitions, improving timing margins and ensuring reliable operation at higher data rates.

Bus Integration

The module is designed for seamless inline insertion into an existing I²C bus and operates transparently without altering the communication protocol or system architecture. It can be positioned at different points along the bus, including near the master device, at the end of the bus, or along extended cable segments, depending on system requirements.

No firmware, configuration, or tuning is required for operation. The module interfaces directly with standard I²C lines and supports flexible integration through dual connectors and a standard pin header, enabling compatibility with both development platforms and embedded systems.

Enable Control

An enable (EN) control pin is provided to allow external control of the active termination function. When the device is enabled, it actively enhances signal transitions on the I²C bus. When disabled or unpowered, the module presents a high-impedance state and does not introduce additional load to the bus.

This behavior ensures compatibility with systems that require selective activation, conditional operation, or power management strategies.

Power and Voltage Compatibility

The module supports a wide supply voltage range, enabling operation across multiple logic-level systems. It is compatible with supply voltages from 1.6 V to 5.5 V and can be used in mixed-voltage I²C environments without requiring additional level-shifting circuitry in typical applications.

This flexibility simplifies system integration and allows the module to be used with a wide variety of microcontrollers, sensors, and embedded platforms.

Low Power Operation

The device incorporates automatic low-power standby functionality to reduce energy consumption during idle conditions. It detects inactivity on the I²C bus and transitions to a low-power state, with shutdown current typically in the microampere range.

This feature makes the module suitable for battery-powered and energy-sensitive applications while maintaining immediate responsiveness during active communication.

Cable Length and Bus Capacitance

The performance advantages of the module become more significant in systems with high bus capacitance. This condition is commonly encountered in long cable runs, multi-device configurations, and distributed sensor networks. Under these circumstances, the module enables reliable operation where conventional I²C communication would typically degrade or fail.

Placement Strategy

For optimal performance, placement near the beginning of the bus is generally recommended. In extended cable applications, positioning the module near critical segments can further improve signal integrity. In complex system topologies, the use of multiple extenders may be considered to maintain consistent performance across the entire bus.

I²C Speed and Performance

The module supports both standard-mode (100 kHz) and fast-mode (400 kHz) I²C communication. It maintains signal quality at higher speeds over extended distances and, in most applications, eliminates the need to reduce clock frequency due to signal degradation.

Compatibility

The module is fully compliant with standard I²C signaling and is compatible with a wide range of devices, including microcontrollers, single-board computers, and sensor modules. It integrates seamlessly with ecosystems such as **STEMMA QT**, **Qwiic** and **DevLab**, enabling rapid prototyping and modular system design.

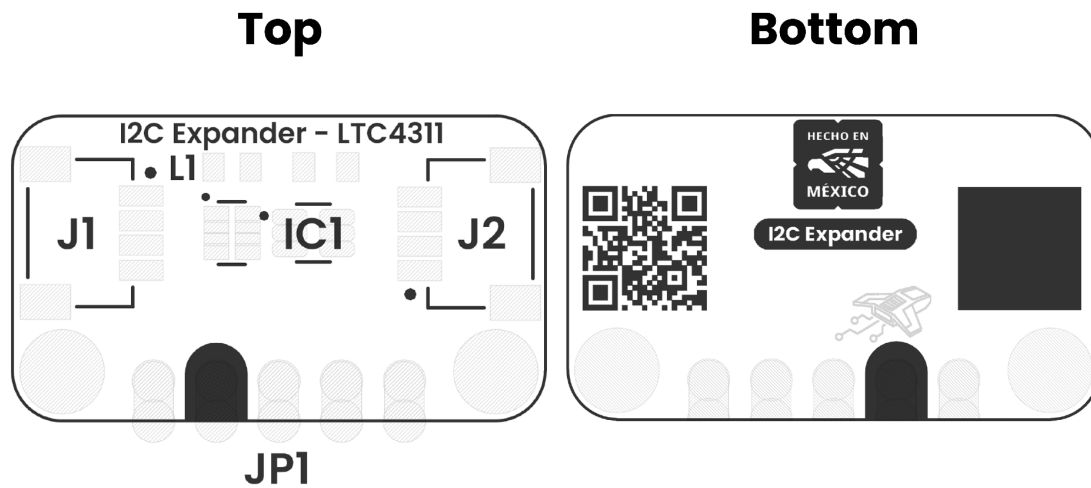
Reliability and Protection

The module is designed to ensure reliable operation in real-world environments. It improves noise margins and enhances signal stability across the I²C bus. Additionally, it incorporates protection against electrostatic discharge (ESD), which refers to the sudden transfer of electrical charge caused by static electricity.

The device is rated to withstand ESD events up to ± 8 kV under the Human Body Model (HBM), a standardized test condition that simulates a person discharging static electricity into the device. This level of protection improves durability during handling, installation, and operation, particularly in applications involving long cables or exposed interfaces.

When disabled, the module does not interfere with normal bus operation, preserving system integrity under all conditions.

3.2 Board Topology



Views of Board Topology

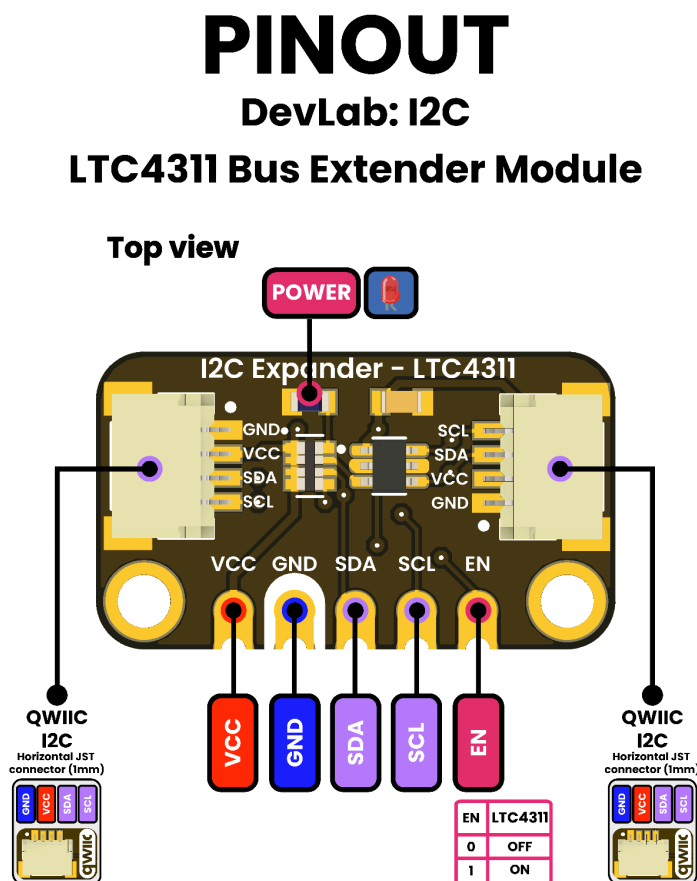
Views of Module Topology

Table 3.2.1 - Components Overview

Ref.	Description
IC1	LTC4311 I2C Extender
L1	Power On LED
JP1	2.54 mm Castellated Holes
J1	QWIIC Connector (JST 1 mm pitch) for I2C
J2	QWIIC Connector (JST 1 mm pitch) for I2C

4 Connectors & Pinouts

4.1 General Pinout



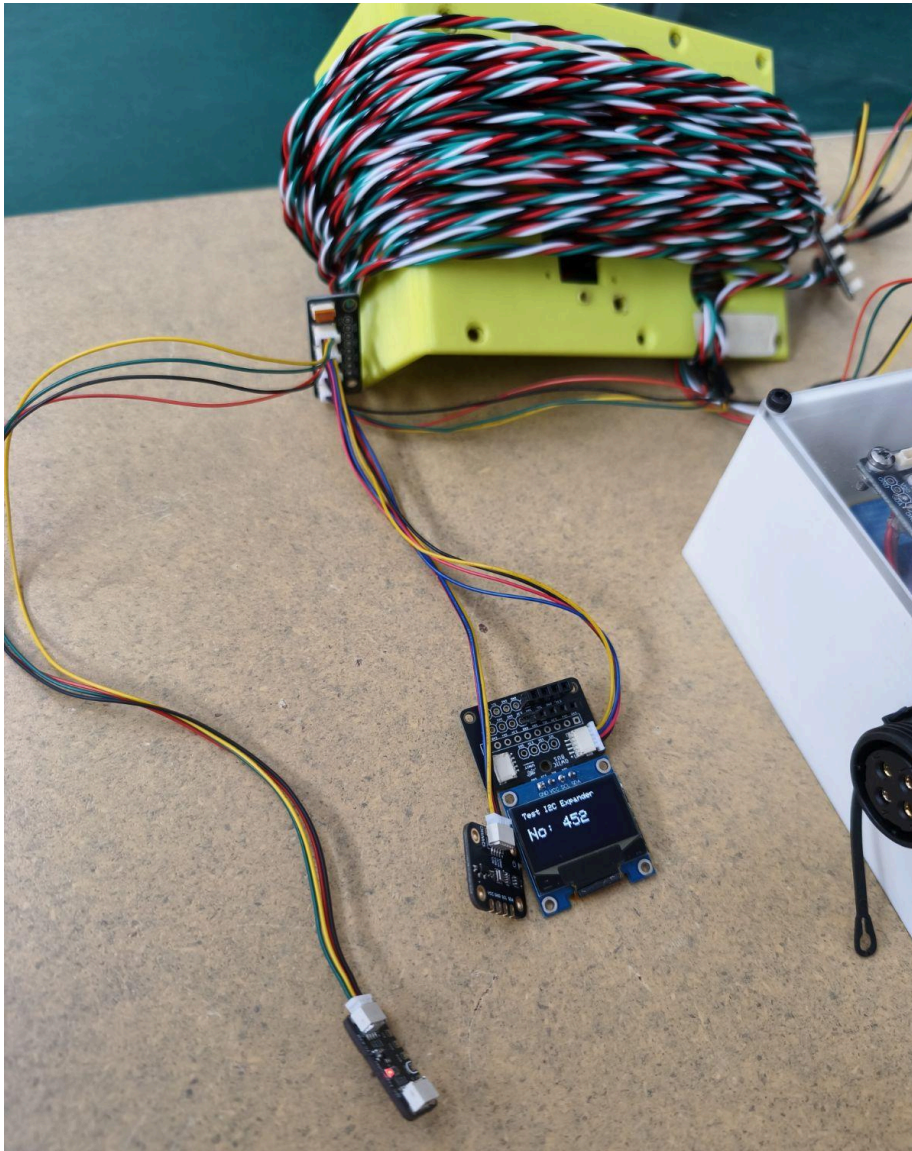
4.2 Pinout General Description

Pin Label	Function	Notes
+3V3	Power Supply	3.3V Power Supply
GND	Ground	Common ground reference
SCL	I2C SCL	Serial Clock Line
SDA	I2C SDA	Serial Data Line
EN	ENABLE	Enables or disables the expander functionality

5 Board Operation

5.1 Getting Started

The I²C LTC4311 Bus Extender Module is designed for immediate integration into an existing I²C system with minimal setup. Proper operation requires only standard I²C connections and a valid supply voltage.



Before powering the system, all connections should be verified. The module must be connected to the I²C bus using the supply voltage (3V3 or within the supported range), ground (GND), serial data line (SDA), and serial clock line (SCL). The enable (EN) pin should be driven high or connected to the supply voltage to activate the device. If the EN pin is held low or left unconnected, depending on system design, the module remains disabled and does not affect the bus.

The module is intended for inline insertion within the I²C bus. It may be placed near the master device, along the communication path, or near the slave devices depending on system topology. For most applications, placement near the beginning of the bus is recommended to maximize signal conditioning effectiveness.

Once powered and enabled, the device operates automatically without requiring firmware configuration, software drivers, or parameter adjustments. It continuously monitors the SDA and SCL lines and enhances signal transitions as needed.

To verify correct operation, the I²C bus can be scanned using the host controller to confirm detection of all connected devices. Stable communication at the desired clock frequency, up to 400 kHz, indicates proper functionality. In systems with extended cabling or multiple devices, improved communication reliability and reduced transmission errors should be observed.

Proper system design should ensure that the bus voltage remains within the specified operating range and that a solid ground reference is maintained. Excessive bus capacitance or improper wiring may still impact performance if system limits are exceeded.

Signal Behavior and Performance

In standard I²C systems, signal rise time is limited by pull-up resistors and total bus capacitance. As cable length increases or additional devices are connected, signal transitions become slower, resulting in rounded waveforms and reduced noise margins. This degradation can limit communication speed and lead to unreliable operation.

When the LTC4311 is powered and enabled, it detects slow rising edges on the SDA and SCL lines and injects controlled current to accelerate these transitions. This process sharpens the waveform and improves signal definition, effectively transforming slow, RC-limited edges into cleaner transitions. The resulting improvement increases timing margins and enables reliable communication at higher data rates.

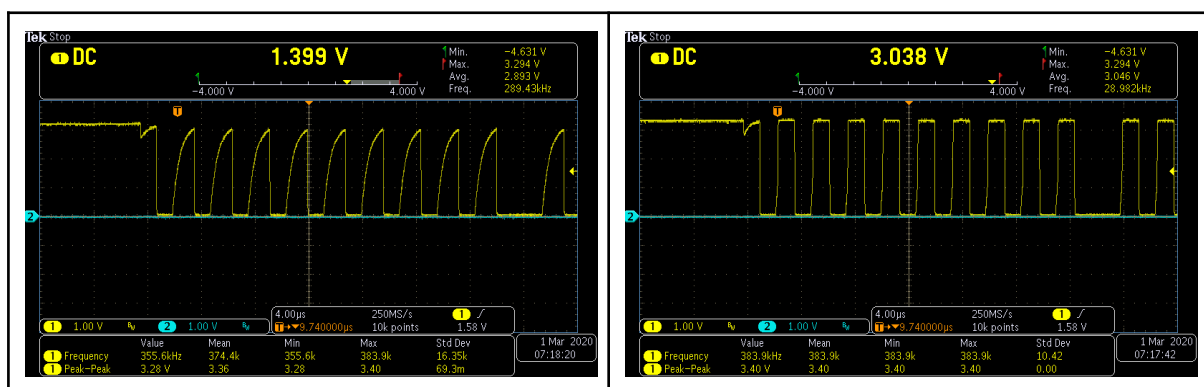


Figure 5-1 illustrates a typical degraded I²C waveform with slow rise time, while Figure 5-2 shows the improved waveform after signal conditioning.

Source: Images adapted from Adafruit, “*Adafruit LTC4311 I²C Extender / Active Terminator*”, available at: <https://learn.adafruit.com/adafruit-ltc4311-i2c-extender-active-terminator>

Operational Principle

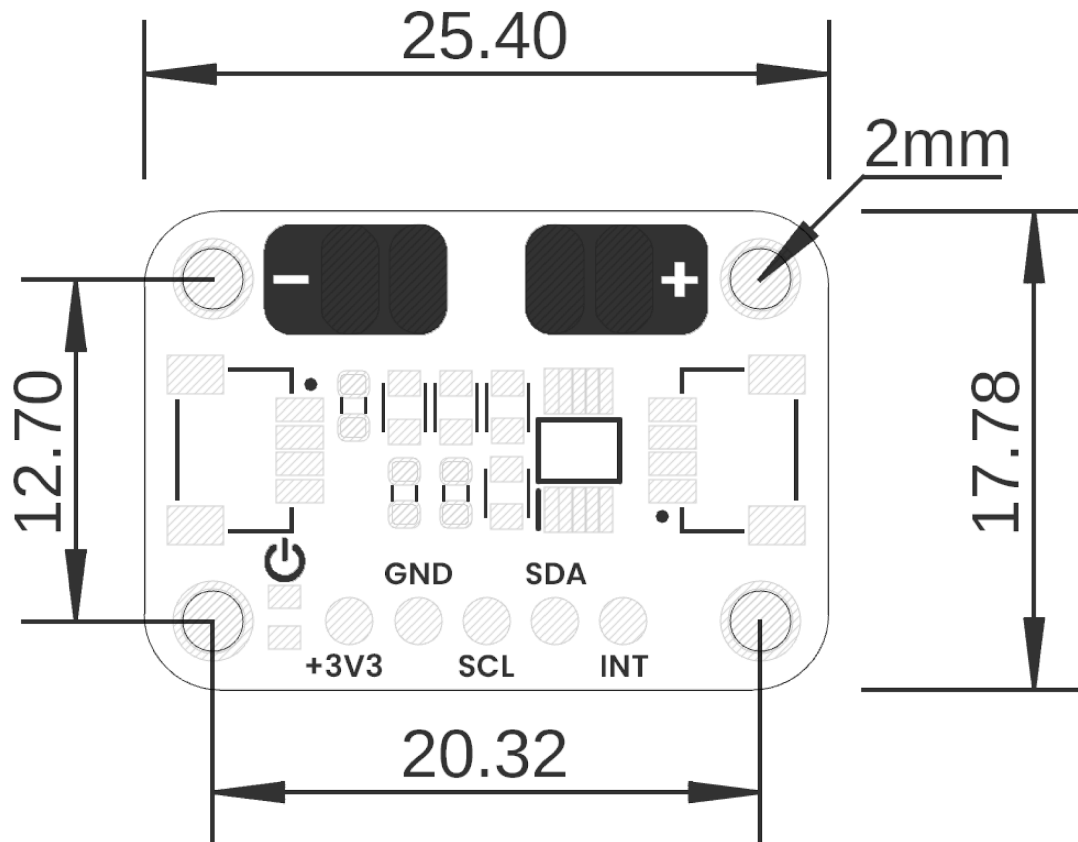
The device operates as an active pull-up accelerator. It remains inactive during steady-state conditions and activates only when a rising edge is detected on the I²C lines. By dynamically assisting the pull-up process, it compensates for the limitations of passive resistor-based designs.

This operation is fully automatic and does not require configuration, calibration, or firmware interaction. The module continuously adapts to the electrical characteristics of the bus in real time, ensuring consistent performance across varying system conditions.

Compatibility and Integration

The module is fully compatible with standard I²C signaling and can be used with a wide range of microcontrollers, single-board computers, and peripheral devices. It supports integration with modular connection ecosystems such as STEMMA QT and Qwiic, enabling rapid development and simplified wiring without the need for soldering.

6 Mechanical Information



Board Dimensions in Millimeters

Mechanical dimensions in millimeters

7 Company Information

Company name	UNIT Electronics
Company website	https://uelectronics.com/
Company Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX

8 Reference Documentation

Ref	Link
Documentation	https://github.com/UNIT-Electronics-MX/unit_devlab_i2c_ltc4311_bus_extender_module
Getting Started Guide	https://wiki.uelectronics.com/wiki/devlab_i2c_ltc4311_bus_extender_module
Adafruit LTC4311 I2C Extender	https://learn.adafruit.com/adafruit-ltc4311-i2c-extender-active-terminator
Thonny IDE	https://thonny.org/
Arduino IDE	https://www.arduino.cc/en/software
Visual Studio Code	https://code.visualstudio.com/download

9 Appendix

9.1 Schematic (https://github.com/UNIT-Electronics-MX/unit_devlab_i2c_ltc4311_bus_extender_module/blob/main/hardware/unit_sch_v_1_0_0_ue0107_i2c_extender.pdf)

